

Mohith Kesavan

Software Engineer with 2+ years of experience building high-throughput backend systems and large-scale geospatial data pipelines. Specialized in Node.js, Python, and PostgreSQL with deep expertise in spatial data migration, ETL architecture, and real-time AI pipelines. Proven track record delivering production systems at scale migrating 8M+ records at 98.6% spatial accuracy and serving 20+ enterprise clients.

mohithkesavan@icloud.com | +91 9361017103 | Coimbatore, Tamil Nadu, India

GITHUB | LINKEDIN | HACKERRANK

EDUCATION

RVS COLLEGE OF ENGINEERING AND TECHNOLOGY, ANNA UNIVERSITY June 2019 - June 2023
B.E. COMPUTER SCIENCE AND ENGINEERING Coimbatore, Tamil Nadu, India
CGPA: 7.8

EXPERIENCE

CREO SYSTEMS PVT. LTD. Coimbatore, Tamil Nadu, India | Feb 2023 – Present

Backend Developer

- Developed analytics reporting features and core modules (audit, ledger) for a Club Management System serving 20+ clients with 500+ players each using Node.js and PostgreSQL.
- Designed DataSync with 1-minute interval synchronization, limiting maximum data loss to 1 minute in failure scenarios and ensuring high availability across all client deployments.
- Implemented asynchronous programming techniques improving system responsiveness and scalability across multiple client deployments within a legacy codebase.

Spatial Data Analyst

- Sole engineer on the Digital Cadastral Modernization (DCM) project designed and delivered a full spatial data migration pipeline over 18 months across 3 feature dumps and 4 landbase dumps.
- Migrated 8,238,462 cadastral features including 7,703,682 anchor points and 802,769 property polygons (11,515,565 coordinates), achieving 98.6% spatial accuracy post-migration.
- Applied linear interpolation for point prediction along cadlines and ST_Translate for Euclidean vector shifting to correct coordinate positions across 7M+ validated records.
- Used azimuth angle calculation to detect and correct angular deviation in line feature alignment during vector shifting.
- Built and executed ETL pipelines across Oracle, PostgreSQL, and SQLite from JSON, CSV, and shapefile sources with property-level attribute and spatial validation throughout.
- Delivered initial deployment and full dataset testing within 4 months, followed by 2 months of incremental processing and pilot area validation using PostGIS, Oracle Spatial, and QGIS.

SKILLS

PROGRAMMING LANGUAGES LIBRARIES/Frameworks

Python, JavaScript, Java, C/C++, Bash, SQL, Node.js
Express.js, Flask, FastAPI, REST APIs, GraphQL, WebSockets, Middleware, OAuth, Authentication, Authorization, Microservices, Redis, Caching, Pandas, NumPy, ETL Pipelines, NLTK, OpenCV, PyTorch, NVIDIA NeMo, CUDA, ASR, TTS, FastConformer, VITS, Llama inference, BeautifulSoup, Selenium, Tkinter, PyPDF, pyttsx3, PyAudio, SoundDevice, MoviePy, FFmpeg, Pillow (PIL), Tesseract 4, Playwright

TOOLS / PLATFORMS

Git, GitHub, Docker, Kubernetes, Linux, Nginx, CI/CD, Postman, GeckoDriver, PostGIS, Oracle Spatial, GeoJSON, GIS, Geospatial Data, Spatial Queries, Co-ordinate Systems, Spatial Analysis, Scalability, Monitoring, Debugging, API Design, Testing, Automation Testing, Logging, Insomnia

DATABASES

Oracle, PostgreSQL, SQLite, MySQL, Redis

PROJECTS / OPEN-SOURCE

REAL-TIME VOICE AI PIPELINE Python, PyTorch, NVIDIA NeMo, FastConformer Hybrid Streaming, VITS, Llama 3.2 1B, NLTK, PyAudio, SoundDevice, CUDA

- Built an end-to-end real-time speech pipeline integrating NeMos FastConformer with cache-aware streaming for ASR at 80ms latency using hybrid CTC+attention decoding.
- Applied NLTK-based sentence segmentation on ASR output to chunk text into semantically coherent units before LLM inference, enabling queue-based processing for efficient memory handling.
- Used VITS as a single end-to-end TTS model handling text-to-waveform in one forward pass, eliminating the traditional acoustic model and vocoder two-stage bottleneck.
- Developed a custom inference loop for Llama 3.2 1B handling tokenization, KV caching, and autoregressive generation without a serving framework, with queued output feeding directly into TTS synthesis.

OCR TEXT EXTRACTION WITH CUSTOM FONT FINE-TUNING Python, Tesseract 4, Pillow (PIL), OpenCV, LSTM

- Built an OCR pipeline fine-tuned on Tesseract 4s LSTM engine for accurate text extraction on stylized typography, specifically targeting the FrenchScriptMT font.
- Generated a custom training dataset of synthetic images using PIL for text rendering and OpenCV for image augmentation simulating real-world variations in lighting, noise, and skew.
- Fine-tuned Tesseract's LSTM model on the generated dataset, producing a domain-adapted OCR engine with recognition accuracy beyond the default models capability on decorative fonts.

AUTOMATED VIDEO SEGMENT EDITOR & EXPORTER Python, OpenCV, FFmpeg, MoviePy, JSON

- Built a video editing pipeline driven by a JSON manifest file defining per-segment configuration for audio handling original audio or dedicated voiceover with manual editing support before final processing.
- Selectively removes or retains audio per segment based on manifest flags, joins all segments into a single continuous video, and applies end normalization at boundaries for artifact-free transitions.
- Supports flexible export with configurable output format and FPS, with voiceover tracks mixed and synchronized over the final video according to manifest-defined mappings.

MULTI-STRATEGY WEB SCRAPER WITH DYNAMIC CONTENT HANDLING Python, Requests, urllib3, wget, BeautifulSoup, Selenium, GeckoDriver, PostgreSQL, JSON

- Built a robust scraping pipeline with multiple HTTP strategies using Requests, urllib3, and wget interchangeably, with timeout detection, configurable retry intervals, and failure fallback handling.
- Integrated Selenium with GeckoDriver to bypass bot detection by simulating a real Firefox browser session for protected or JS-heavy pages with load-wait logic for dynamic content.
- Used BeautifulSoup for structured HTML parsing with support for automated element resolution and manual CSS/XPath path entry for targeted value extraction.
- Persisted scraped data to PostgreSQL with the pipeline scheduled nightly at 12AM for consistent daily updates.

TTS PDF READER WITH PLAYBACK CONTROLS Python, Tkinter, PyPDF, NVIDIA NeMo, VITS, pyttsx3, Microsoft Neural Voices

- Built a desktop PDF reader with a Tkinter UI extracting text via PyPDF and synthesizing speech using NVIDIA NeMos VITS model for high-quality neural TTS output.
- Integrated pyttsx3 with Microsoft Neural Voices as an alternative engine, giving users voice selection flexibility.
- Implemented playback controls including adjustable speed and per-page selection, allowing navigation to specific sections without linear playback.

CERTIFICATIONS

- Information and Communication Technology - Tamil Nadu Newsprint and Papers Limited.
- SQL (Advanced) Certificate - **HACKERRANK**

HONORS & AWARDS

- Student President, Department of Computer Science and Engineering Incubation Cell, RVS College of Engineering and Technology (2021-2023)
- Student Secretary, Department of Computer Science and Engineering, RVS College of Engineering and Technology (2021-2023)